Responsible Machine Learning
Assignment 7
40 points

November 6, 2026

In Assignment 7, work alone or with a group to create a model card for your best model from previous assignments. You may use AI tools, but you are responsible for validating and understanding your work.

1 Recommended References and Examples.

Model cards document a model's intended uses, development, evaluation, and limitations. They help data scientists communicate about models so that others can maintain them and respond to incidents. Model documentation can also support executive review and may be required by regulation.

Recommended references:

- *Model Cards for Model Reporting*
- Model Cards (Google DeepMind)

Example: `https://github.com/jphall663/GWU_DNSC_6301_project`. This example is incomplete but may help you get started.

2 Instructions and Rubric.

Keep the model card concise. You may use bullet points, charts, and tables; number and describe each chart or table. The model card should focus on your best-performing model from previous assignments. Discuss additional models only as points of comparison, and explain why you selected the primary model.

Rubric:

- Structure (12 pts.)
  - Name and contact information for yourself or all group members. You may anonymize this information for the assignment. (2 pts.)
  - Clearly delineated sections for:
    - Intended use (1 pt.)
    - Training data (1 pt.)
    - Evaluation data (1 pt.)
    - Model details (1 pt.)
    - Quantitative analysis (1 pt.)
    - Ethical considerations (1 pt.)
  - Correct and informative external links, such as links to training and evaluation data (1 pt.)
  - Correct use and introduction of abbreviations (1 pt.)

  - Meaningful and informative charts or tables (1 pt.)
  - Proper descriptions of charts or tables (1 pt.)

- Content (26 pts.)
  - Intended use (4 pts.)
    * Describe the business value of your best model.
    * Describe how the model is designed to be used and identify its intended users.
    * State whether the model can be used for additional purposes.
  - Training data (4 pts.)
    * State the source of the training data and how it was divided into training and validation data.
    * State the number of rows in the training and validation data.
    * Define all training data columns and any engineered columns.
  - Evaluation data (4 pts.)
    * State the source and number of rows of the evaluation (or test) data.
    * Describe any differences in columns between the training and evaluation data.
  - Model details (4 pts.)
    * State the input and target columns used by your best model.
    * State the model type, software and software version, and the model's hyperparameters or other settings.
  - Quantitative analysis (6 pts.)
    * State the metrics used to evaluate your best model (1 pt.).
    * Report these metrics for the training, validation, and evaluation (or test) data (1 pt.).
    * Submit your predictions to the class evaluation server at a TBD URL and report your model's leaderboard rank (1 pt.).
    * Include at least one plot or table from each weekly assignment, for a total of at least six (1 pt.).
    * Include global variable importance and partial dependence for your best model (1 pt.).
    * Briefly compare additional models as alternatives and explain why you selected your best model (1 pt.).
  - Ethical considerations (4 pts.)
    * Describe potential negative impacts and uncertainties related to using your best model.
    * Consider math or software problems and real-world risks, including who could be affected, what could happen, and when and how it might occur.
    * Describe any unexpected results encountered during training.

- Grammar and spelling (2 pts.)
  - Correct grammar and punctuation (1 pt.)
  - Correct spelling (1 pt.)

3 Submit Code and Model Card.

Put your model card in the `README.md` file of a public GitHub repository, or a private repository shared with the instructor (`jphall663`). Submit your model card to the class evaluation server at a TBD URL.

Your deliverables are due: TBD.